



Space Technology and Applications

Course code: ET375TGL

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Unit-2

Satellite Sub-Systems: Classification of satellites., Payloads, Subsystems-Altitude and orbit control system, TT&C Sub-System, Power Systems.

Communication sub systems: Transponder, Satellite antennas and Parameters, Satellite Antennas in Practice



Classification of Satellites

Spacecrafts are classified based on

- Orbit,
- Service,
- Size



Classification by Orbital Altitude

- Geostationary Earth Orbit (GEO) / Geostationary Satellite Orbit (GSO)
- Low Earth Orbit (LEO)
- Medium Earth Orbit (MEO)



Classification by Orbital Characteristics

- Polar Orbiting Satellites
- Sun Synchronous Orbit (SSO)
- Highly Elliptical Orbits (HEO)



Classification by Mission or Service

- **Fixed Satellite Service (FSS)**
 - Example: Point to Point Communication
- **Broadcasting Satellite Service (BSS)**
 - Example: Satellite Television/Radio
 - Also called Direct Broadcast Service (DBS).
- **Mobile Satellite Services (MSS)**
 - Example: Satellite Phones
- **Navigational Satellite Services**
 - Global Positioning System (GPS)
- **Meteorological Satellite Services**
 - monitoring and reporting weather conditions
- **Search and Rescue (SAR)**
- **Earth Observation/Remote Sensing**
 - detection of water pollution, agricultural surveys, and military surveillance



Classification by Size and Power

- Classification by Size (Mass/Volume)

- Traditional GEO Satellites
- Small Satellites (SmallSats)
 - Cubesats
 - MicroSats, NanoSats, PicoSats

- Classification by Power

- High power
- Medium power
- Low power



Satellite Subsystems

- Attitude and Orbit Control System (AOCS)
- Telemetry, Tracking, Command, and Monitoring (TTC&M)
- Power System
- Communications Subsystems
- Satellite Antennas

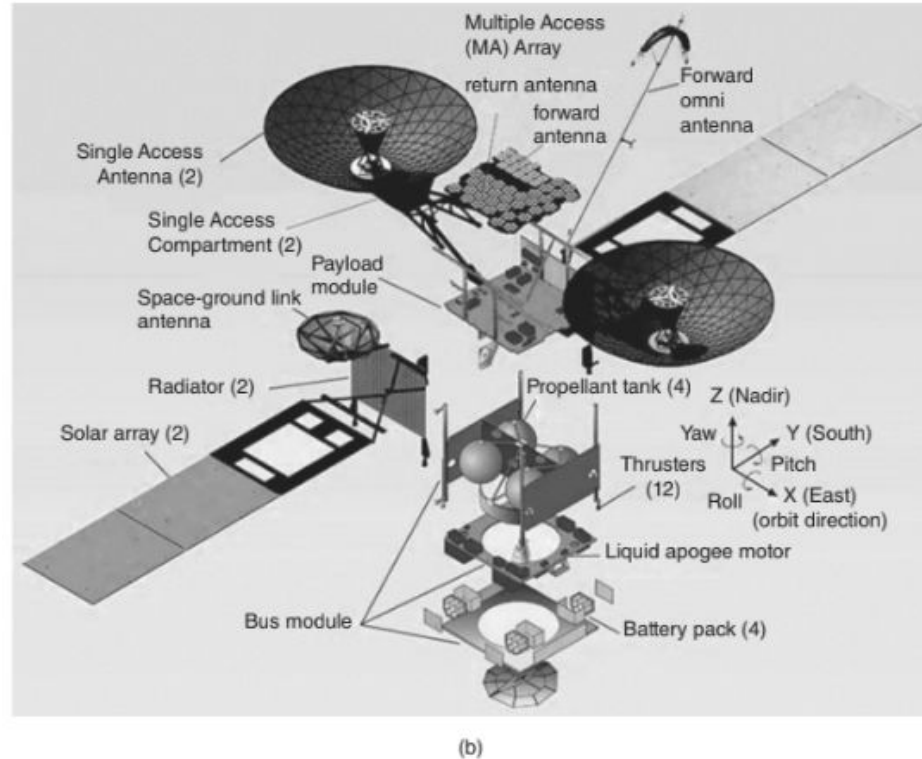
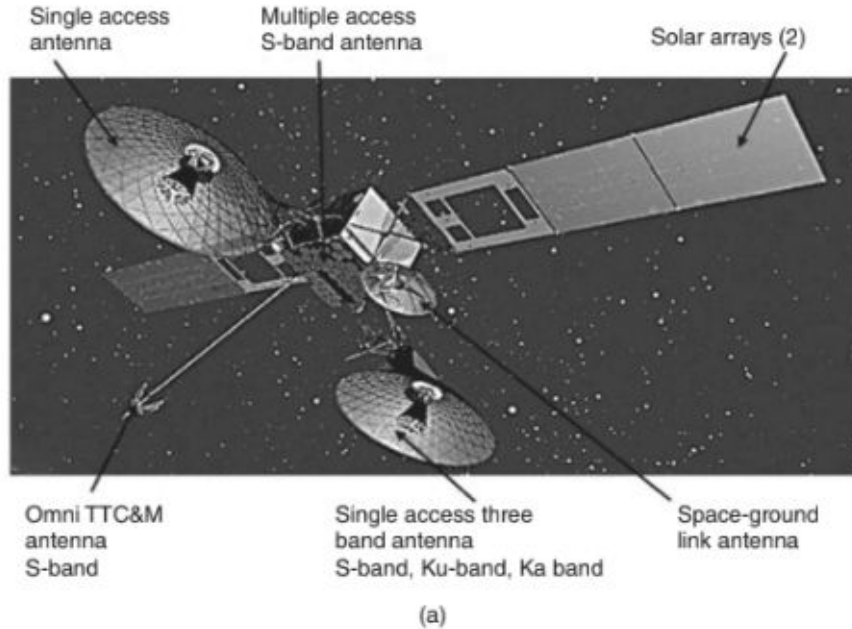


Figure 3.1 NASA third generation TDRS satellite. (a) TDRS satellite in orbit. (b) Exploded view of TDRS satellite. The satellite is inverted from Figure 3.1a. Source: NASA.



Payload

The basic payload in the case of a communication satellite, for instance, is a transponder, which acts as a receiver/amplifier/transmitter. A transponder can be considered to be a microwave relay channel that also performs the function of frequency translation from the uplink frequency to the relatively lower downlink frequency.

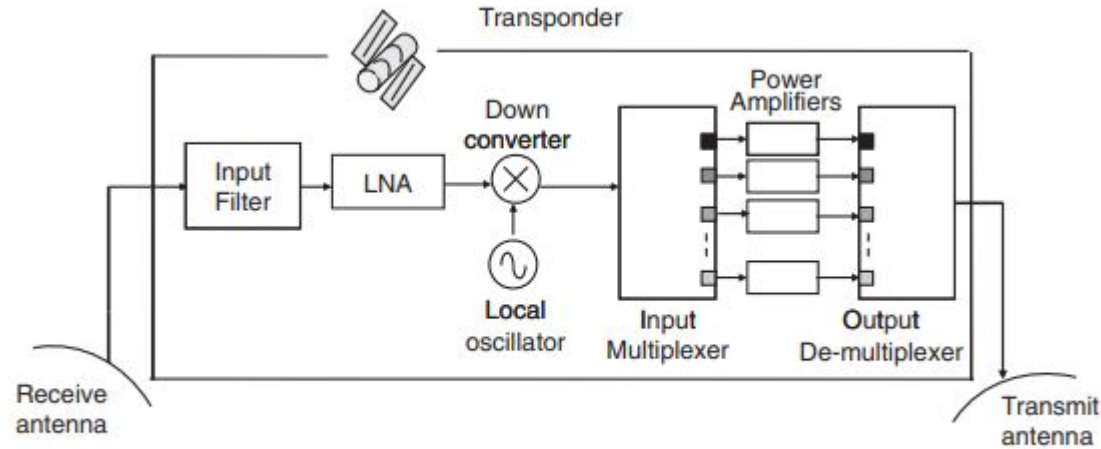


Figure 4.22 Transponder



- In the case of weather forecasting satellites, the radiometer is the most important payload. The radiometer is used as a camera and has a set of detectors sensitive to the radiation in the visible, near-IR and far-IR bands.
- High resolution visible (HRV) cameras, multispectral scanners and thematic mapper are the main payloads on board an Earth observation satellite.



Attitude and Orbit Control

- The attitude and orbit control subsystem performs twin functions of controlling the orbital path, which is required to ensure that the satellite is in the correct location in space to provide the intended services and to provide attitude control, which is essential to prevent the satellite from tumbling in space.
- In addition, it also ensures that the antennae remain pointed at a fixed point on the Earth's surface.
- The requirements on the attitude and orbit control subsystem differ during the launch phase and the operational phase of the satellite.



Attitude Control

- Attitude of a satellite, or for that matter any space vehicle, is its orientation as determined by the relationship between its axes (yaw, pitch and roll) and some reference plane.
- The attitude control subsystem is used to maintain a certain attitude of the satellite, both when it is moving in its orbit and also during its launch phase.
- Two types of attitude control systems are in common use, namely spin stabilization and three-axis stabilization.
- During the launch phase, the attitude control system maintains the correct attitude of the satellite so that it is able to maintain link with the ground Earth station and controls its orientation such that the satellite is in the correct direction for an orbital manoeuvre.



- When the satellite is in orbit, the attitude control system maintains the antenna of the satellite pointed accurately in the desired direction
- Attitude control systems can be either passive or active.
- Passive systems maintain the satellite attitude by obtaining equilibrium at the desired orientation. There is no feedback mechanism to check the orientation of the satellite.
- Active control maintains the satellite attitude by sensing its orientation along the three axes and making corrections based on these measurements.
- The basic active attitude control system has three components: one that senses the current attitude of the platform, second that computes the deviations in the current attitude from the desired attitude and third that controls and corrects the computed

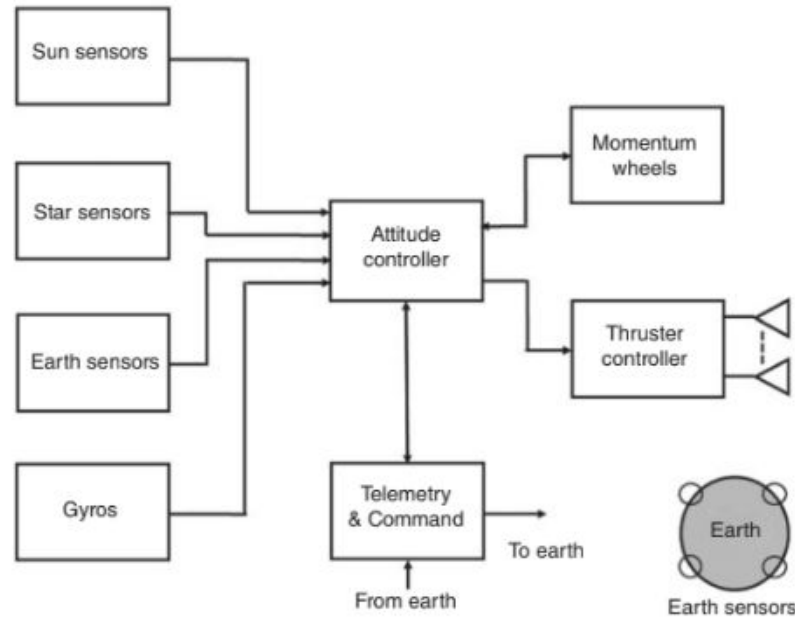


Figure 3.5 Attitude control system for a three-axis stabilized GEO satellite. The sun and star sensors are used to establish orientation with respect to space and the earth sensors are used for z-axis pointing. The gyros detect movement on the three axes of the satellite and send signals to the controller to initiate a correction with the momentum wheels. The thrusters used for attitude control are ion thrusters. The attitude controller operates autonomously and sends all sensor data and command data to the controlling earth station over the telemetry link.



Orbit Control

- Orbit control is required in order to correct for the effects of perturbation forces.
- These perturbation forces may alter one or more of the orbital parameters.
- The orbit control subsystem provides correction of these undesired changes. This is usually done by firing thrusters.
- During the launch phase, the orbit control system is used to affect some of the major orbit manoeuvres and to move the satellite to the desired location.

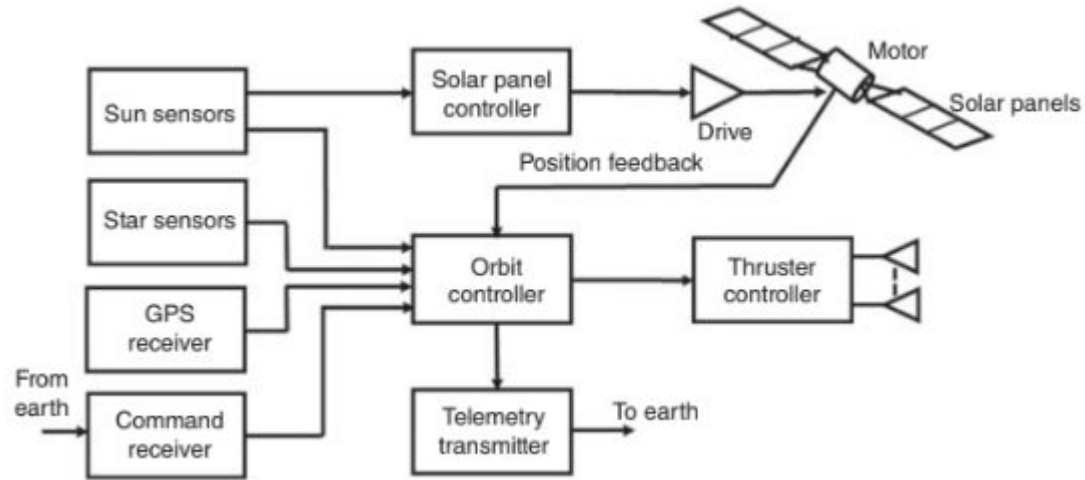


Figure 3.7 Simplified orbital control system.



Tracking, Telemetry and Command Subsystem

- The tracking, telemetry and command (TT&C) subsystem monitors and controls the satellite right from the lift-off stage to the end of its operational life in space.
- The tracking part of the subsystem determines the position of the spacecraft and follows its travel using angle, range and velocity information.
- The telemetry part gathers information on the health of various subsystems of the satellite.
- It encodes this information and then transmits the same towards the Earth control centre.
- The command element receives and executes remote control commands from the control centre on Earth to effect changes to the platform functions, configuration, position and velocity.
- The TT&C subsystem is therefore very important, not only during orbital injection and the positioning phase but also throughout the operational life of the satellite.

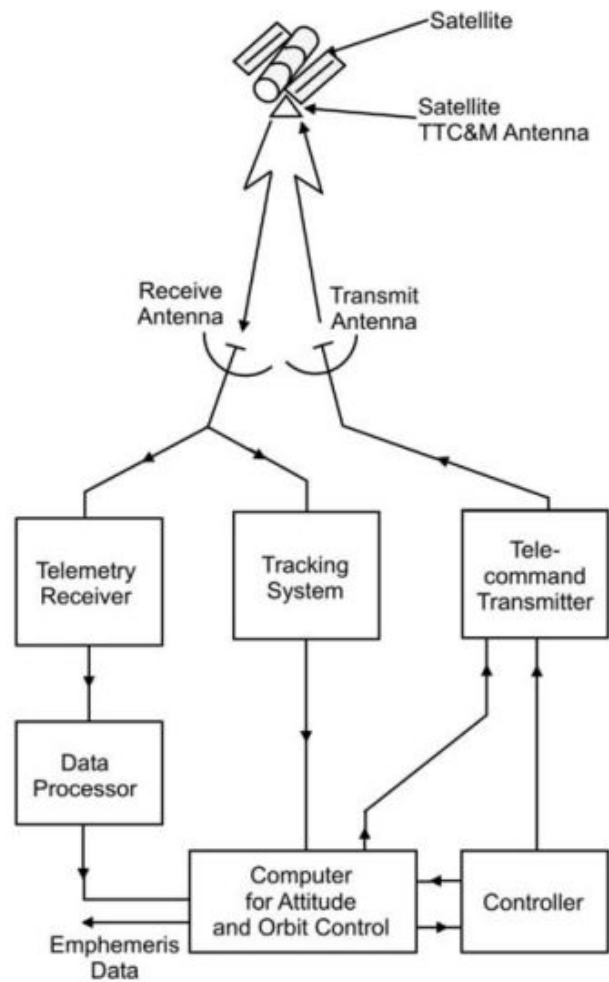


Figure 4.21 Block schematic arrangement of the basic TT&C subsystem



- TTC&M is an essential system for the successful operation of a communications satellite
- The systems are partly on the satellite and partly at the controlling earth station
- The TTC&M process is crucial to the satellite management task, involving a dedicated earth station and personnel
- Earth stations used for orbit determination and range measurements are generally referred to as TTC&M stations
- Main Functions of Satellite Management:
 - Control the orbit and attitude of the satellite (AOCS)
 - Monitor the status (health) of all sensors and subsystems on the satellite
 - Switch on or off sections of the communication system

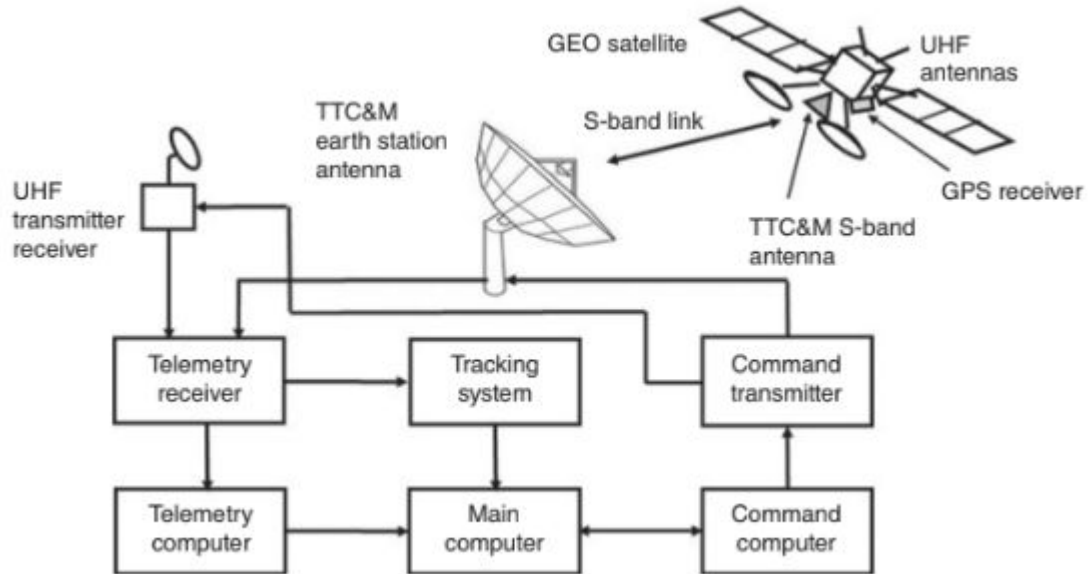


Figure 3.8 Simplified diagram of the earth based control system for a GEO satellite.



Telemetry and Monitoring System

- **Function:** The monitoring system collects data from many sensors within the satellite and sends this data back to the controlling earth station via the telemetry link
- **Data Collected (Satellite Health)**
 - Pressure in fuel tanks
 - Voltage and current in the power conditioning unit
 - Critical voltages and currents in the communications electronics
 - Temperature of many subsystems (must be kept within predetermined limits)
 - Status of subsystems and positions of switches
 - Sighting devices used to maintain attitude are also monitored



- **Data Transmission:**

- Telemetry data is usually digitized
- Transmitted as Phase Shift Keying (PSK) of a low-power telemetry carrier using Time Division Multiplexing (TDM)
- A low data rate is used to allow the receiver to have a narrow bandwidth and maintain a high Carrier to Noise Ratio (CNR)
- A computer at the controlling earth station is used to monitor, store, and decode the telemetry data



Tracking

- **Function:** The tracking system is located at the earth station and provides information on the range, elevation, and azimuth angles of the satellite
- **Orbit Determination:**
 - Repeated measurement of these three parameters permits computation of orbital elements
 - This allows changes in the satellite's orbit to be detected
- **Tracking Methods Used:**
 - **Doppler Shift:** The controlling earth station can observe the Doppler shift of the telemetry or beacon carrier to determine the rate at which range is changing
 - **Active Range Determination:** Achieved by transmitting a pulse or sequence of pulses to the satellite and observing the time delay before the pulse is received again (propagation delay in the transponder must be accurately known)
 - **Triangulation:** If a sufficient number of earth stations make simultaneous range measurements, the satellite's position can be determined within 10 m
 - **Onboard GPS:** Some satellites carry GPS receivers that report the satellite's position over the telemetry link



Command System

- Function: The command system is vital for successful launch and operation, used to make changes in attitude, correct the orbit, and control the communication system
- During launch, it is used to control the firing of the Apogee Kick Motor (AKM) and to extend the solar arrays and antennas
- Security and Validation:
 - The command structure must have safeguards against unauthorized attempts or inadvertent operation
 - Encryption of commands and responses is used to provide security (e.g., Intelsat VI and all subsequent Intelsat satellites used encryption)
 - To minimize the risk of erroneous commands: the command word is sent, the satellite checks validity and sends it back via telemetry, the earth station verifies it, and then sends an execute instruction



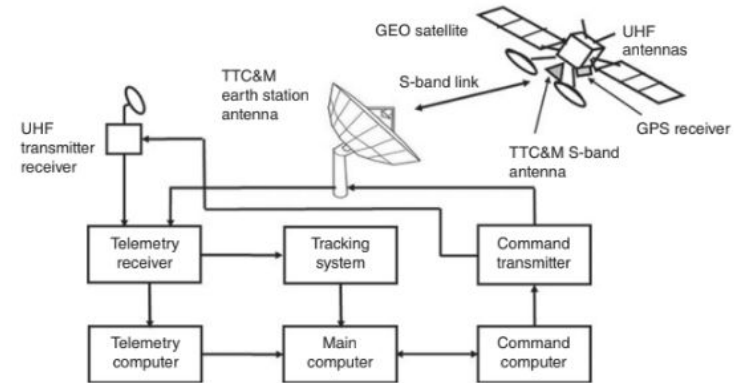
TTC&M Communication Links

- **Primary Links (On-Station):**
 - The command and telemetry links are usually separate from the main communication system on a GEO satellite.
 - They may operate in the same frequency band, often C-band (6 and 4 GHz), utilizing a gap between the communication channel frequencies.
 - TDRS satellites use S-band for TTC&M
- **Backup System (Launch/Emergency):**
 - A backup system is essential during the launch phase when the main TTC&M system may be inoperable (e.g., if the satellite does not have the correct attitude).
 - This backup uses near omnidirectional antennas at Ultra High Frequency (UHF) or S-band (2–4 GHz).
 - It provides control of critical functions (AKM, attitude/orbit thrusters, power conditioning unit).
 - In the event of failure of the main system, the backup can keep the satellite on station.
 - It is also used to eject the satellite from geostationary orbit and switch off all transmitters at the end of its useful life



TTC&M Earth Station Functions

- Controlling Earth Station (Figure 3.8):
 - The TTC&M earth station may be owned by the satellite owner or contracted out by a third party.
 - It is equipped with antennas for S-band (on-station) and potentially UHF (launch phase) links.
 - The main computer receives telemetry data from the Attitude and Orbital Control System (AOCS) and numerous sensors.
 - The computer stores telemetry data for later analysis to determine aging trends in critical components.
 - The command computer translates commands for station keeping and switch settings into command codes.
 - The earth station also performs tracking, measuring angular position and range.





Power Systems

- All communications satellites obtain their electrical power from solar cells, which convert incident sunlight into electrical energy.
- Power is necessary for both the communications payload (transmitters) and all other electrical systems, referred to as housekeeping.
- Power systems generate DC power ranging from as little as 1 W (for a 1U cubesat) up to 20 kW (for a large GEO platform).
- **Note:** Thermonuclear generators are generally not used on communications satellites due to the danger of failed launches spreading nuclear fuel, although they are used on deep space probes where sunlight intensity is too low (e.g., near Jupiter, where intensity is 1% of Earth orbit)



Solar Power Generation

- Solar Energy Input:
 - • In the vacuum of outer space at geostationary altitude, the sun's radiation intensity is 1.36 kW/m^2
- Solar Cell Technology and Efficiency:
 - Gallium arsenide (GaAs) cells are typically used, exhibiting efficiencies of 33–39% at the beginning of life (BOL).
 - By contrast, typical terrestrial silicon cells have 10–19% efficiency.
 - Efficiency decreases over time due to aging and micrometeor impacts.
 - To ensure sufficient power at the end of the satellite's life (EOL), approximately 15% extra solar cell area is typically provided



Solar Array Design and Mechanics (GEO Satellites)

- **Deployment on Three-Axis Stabilized Satellites:**
 - • Solar cells are mounted on flat panels (sails) typically running along the Y-axis of the satellite.
 - • These panels are rotated by an electric motor to maintain normal incidence of the sunlight.
- **Operational Challenges:**
 - **Heating and Voltage Drop:** Maintaining normal incidence causes the cells to heat up, typically reaching 50–80° C, which results in a drop in output voltage.
 - **Current Transfer:** A rotary joint with slip rings is required to safely transfer the high electrical current from the rotating sails to the satellite body.
 - **Failure Risk:** For large GEO satellites generating 20 kW, these slip rings may carry currents up to 200 A (at 50 V bus voltage), and failures have occurred, which can cut the available power in half.



Battery Systems and Eclipse Operations

- **Role of Batteries:**
 - Batteries are essential to supply power to all subsystems during launch and during eclipses.
 - Eclipses occur twice annually, coinciding with the spring and fall equinoxes.
 - The longest eclipse duration is 70 minutes.
- **Battery Management:**
 - Batteries must be designed to operate within a specified maximum depth of discharge. If discharged below this limit, they may not recover full operational capacity after recharging.
 - Nickel-hydrogen batteries were traditionally favored, operating around 70% depth of discharge.
 - Lithium-ion batteries are now widely used due to their higher capacity per unit weight.
 - Typical battery voltages range from 20–50 V with capacities of 20–1000 ampere hours (AHs).



Power Conditioning and Monitoring

- **Power Conditioning Unit (PCU):**
 - This unit controls the charging current supplied by the solar cells.
 - It is also responsible for dumping excess current from the solar cells into specific devices like heaters or load resistors on the cold side of the satellite.
- **Monitoring (via Telemetry):**
 - The power subsystem is continuously monitored by sensors.
 - Sensors track temperature, voltage, and current on the batteries, power regulator, and solar cells.
 - This data is supplied to the onboard control system and transmitted back to the controlling earth station via the telemetry downlink.



Communications Subsystems

The fundamental purpose of a communications satellite is to provide a platform in orbit for the relaying of voice, video, and data communications.

- All other subsystems on the satellite (AOCS, power, TTC&M, etc.) exist solely to support the communications system,.
- Communications satellites are designed to provide the largest traffic capacity possible.
- Successive generations of GEO satellites have drastically increased capacity, resulting in a lower cost per transmitted bit.
 - Large GEO satellites today can carry up to 140 Gbps of data.

Key Components:

- Antennas: Two or more antennas for receiving and transmitting signals over wide bandwidths.
- Transponders: A set of receivers and transmitters that amplify and retransmit incoming signals.



Transponders: Function and Types

A transponder is a device that receives radio signals and retransmits them, typically at a different frequency.

- **Linear (Bent Pipe) Transponders**
 - This type simply amplifies the received signal and changes its frequency for retransmission,,.
 - The analogy is to a pipe that conveys water from one point to another.
 - Bent pipe transponders can carry analog or digital signals,.
 - They provide flexibility, as they can be used for any combination of signals that fits within their bandwidth,.
- **Baseband Processing (OBP) Transponders**
 - Used only with digital signals,.
 - Converts the received signal to baseband, processes it, and then remodulates and retransmits a digital signal,.
 - Enables complex operations like extracting packet headers for routing and switching,.
 - OBP essentially creates a router in the sky.



Transponder Architecture and Frequency Conversion

Frequency Conversion (Down Conversion):

- A bent pipe transponder uses a mixer and a local oscillator (LO) to change the uplink frequency (e.g., 6 GHz) to the downlink frequency (e.g., 4 GHz),.
- The frequency shift (e.g., 2225 MHz for 6/4 GHz links) ensures the downlink signal is different from the uplink signal.
- Double conversion is used in some bands (e.g., 14/11 GHz), where the signal is first converted to an Intermediate Frequency (IF) before being up-converted to the final downlink frequency.

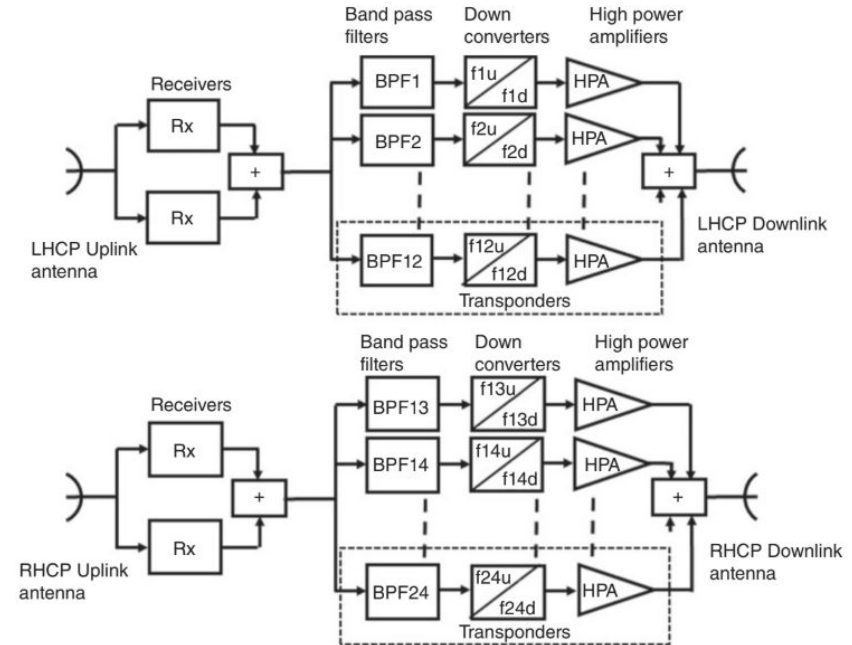
Channelization:

- The allocated bandwidth (e.g., 500 MHz in C-band) is divided into multiple channels (e.g., twelve 36 MHz wide channels), each handled by a separate transponder.

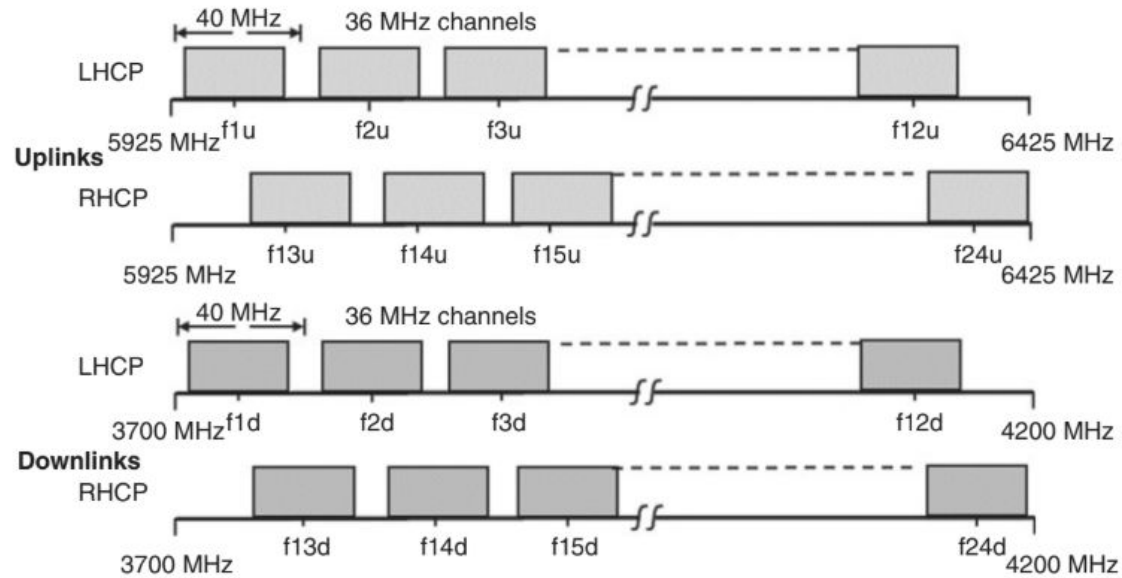


Maximizing Capacity: Frequency Reuse

- Frequency reuse is a key technique to increase capacity by utilizing the same RF spectrum multiple times,.
- Orthogonal Polarization:
 - Capacity is doubled by carrying communication signals on two separate polarizations (e.g., Left Hand Circularly Polarized (LHCP) and Right Hand Circularly Polarized (RHCP)),,.
 - A 20 MHz frequency offset between orthogonally polarized transponders helps receivers separate the signals and reduces crosstalk.



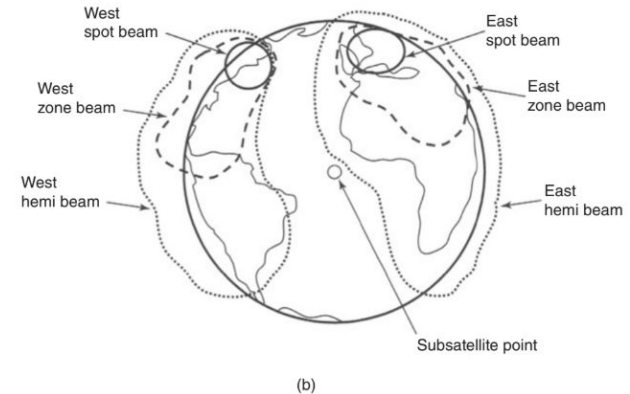
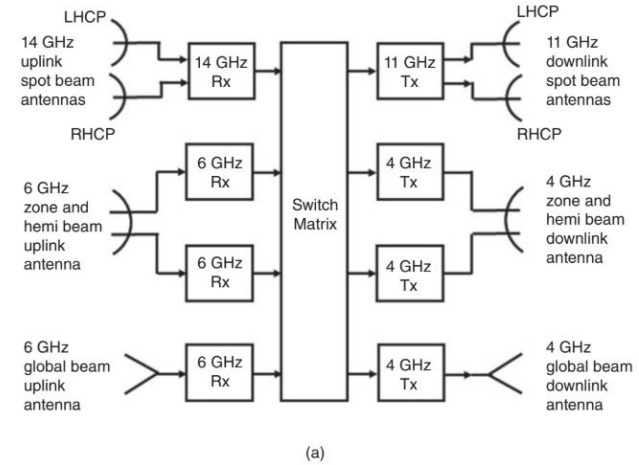
Example of a system of 24 transponders for a 6/4 GHz satellite operated with orthogonal circularly polarized signals.

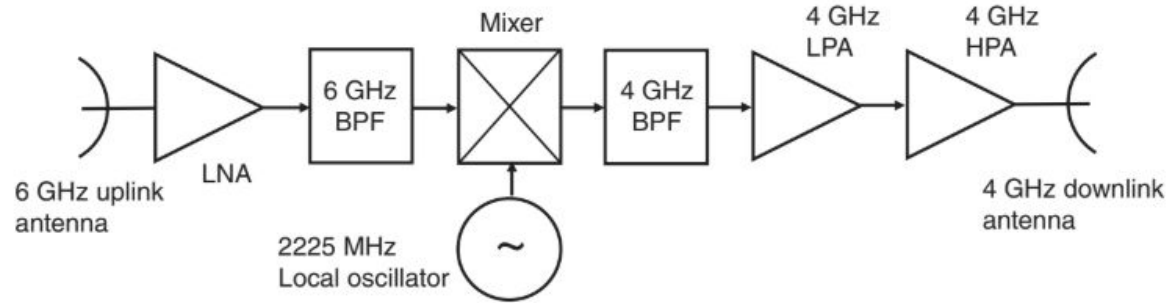




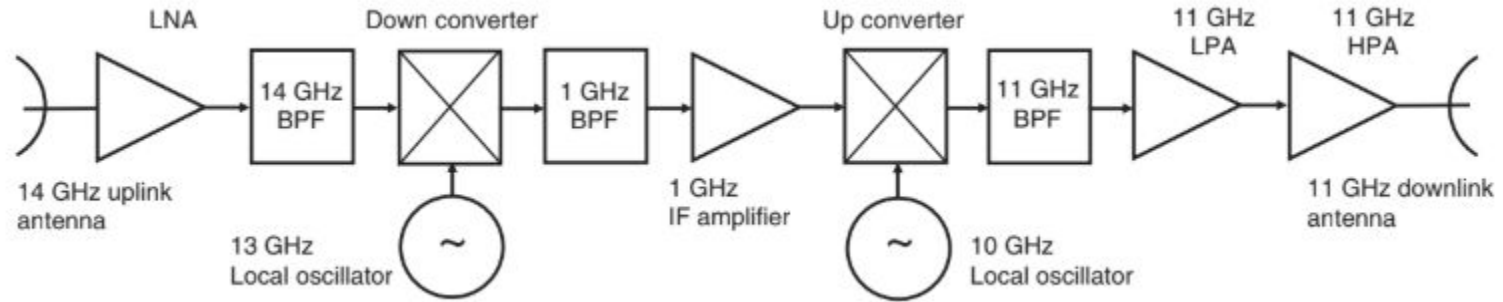
Multiple Beams (Spot Beams):

- The communication capacity is increased by using spot beam antennas that direct multiple narrow beams at specific areas,.
- Because the beams are spatially separated, the same RF frequencies can be reused many times,.
- The combination of dual polarization and multiple spot beams can achieve high efficiency, such as 18-fold frequency reuse achieved on the ViaSat-I satellite in the 30/20 GHz band.

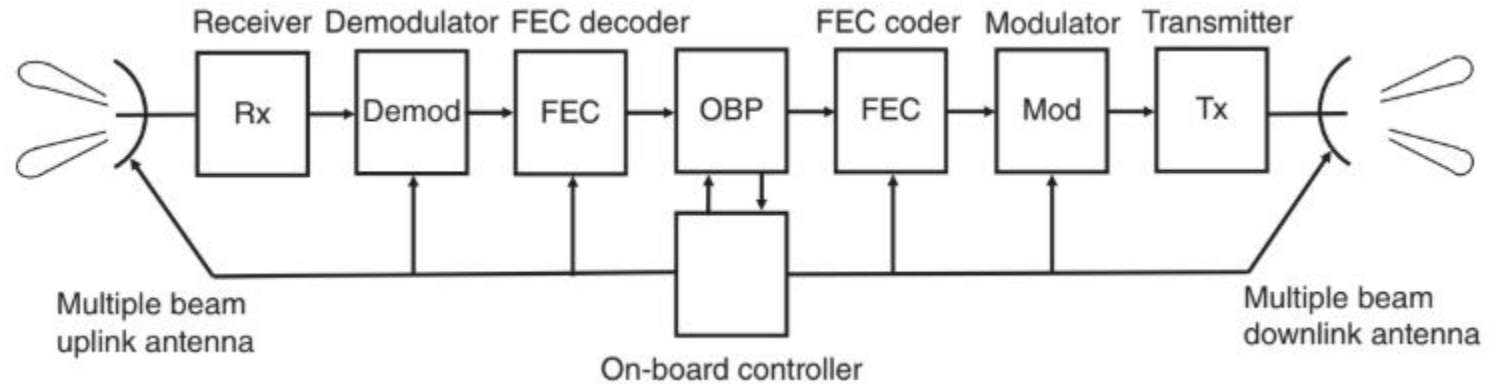




Simplified diagram of a bent pipe transponder for the 6/4 GHz band.



Double frequency conversion bent pipe transponder for the 14/11 GHz band.



Onboard processing (OBP) transponder with multiple beam antennas.



Antenna Subsystem

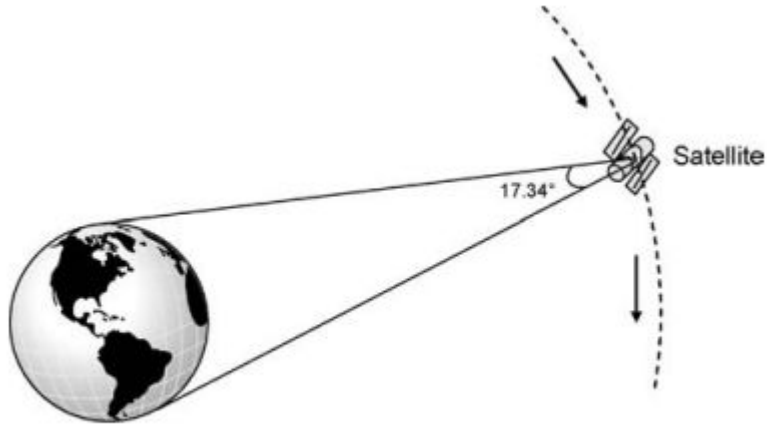


Figure 4.23 Beam width of a global antenna



Figure 4.25 Parabolic reflector antenna (Reproduced by permission of © 3amSystems)



Figure 4.24 Intelsat-4 (Reproduced by permission of © Intelsat)



Figure 4.26 Horn antenna (Reproduced by permission of © A.H. Systems Inc.)



Figure 4.27 Omnidirectional antenna (Reproduced by permission of © Dudley Lab)



Antenna Parameters

1. Gain
2. Effective isotropic radiated power (EIRP)
3. Beam width
4. Bandwidth
5. Polarization
6. Aperture



Types of Antennas

- Reflector Antennas

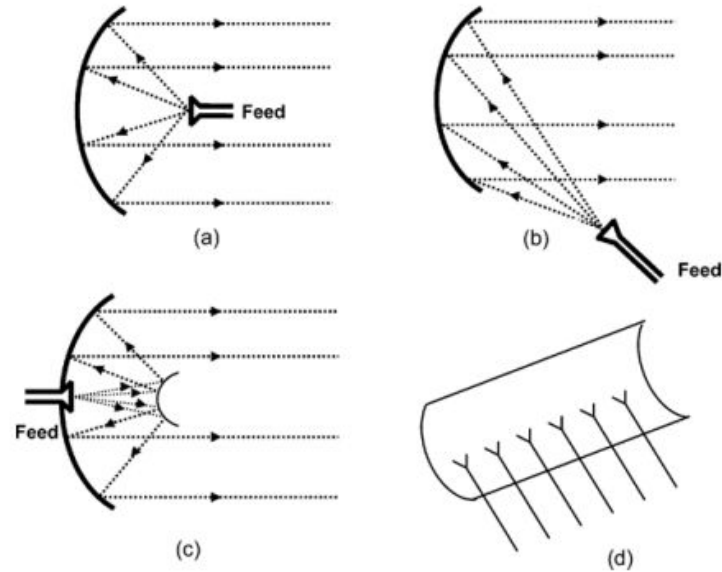


Figure 4.30 (a) Focal point fed parabolic reflector, (b) offset fed sectioned parabolic reflector, (c) cassegrain fed reflector and (d) array fed cylindrical reflector



Horn Antennas

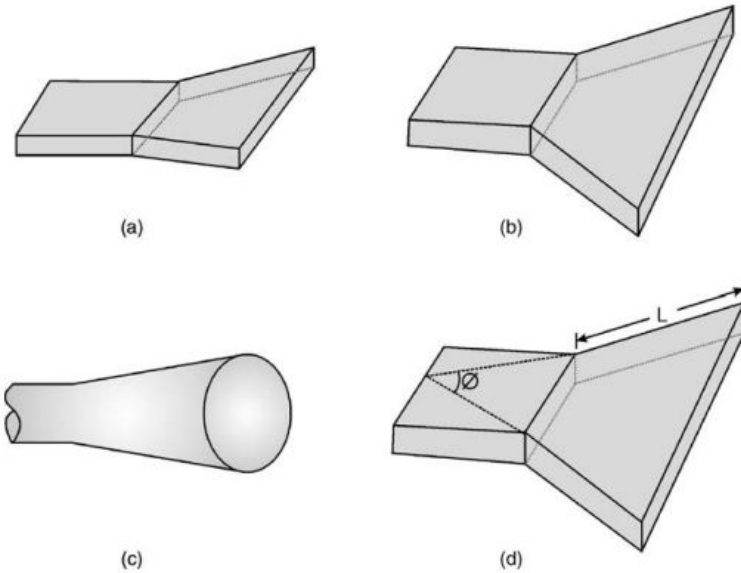


Figure 4.34 (a) Sectoral horn antenna, (b) rectangular pyramidal horn antenna and (c) conical horn antenna. (d) important design parameters of a horn antenna

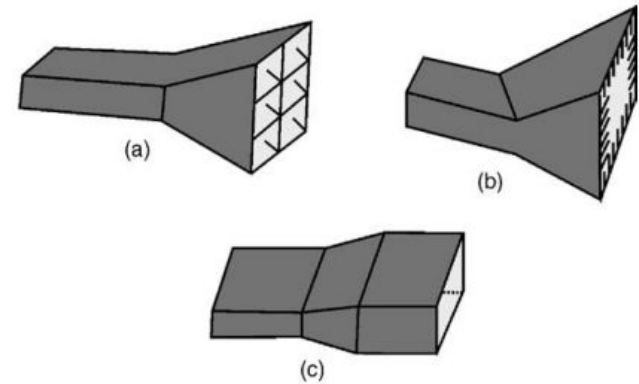


Figure 4.35 (a) Segmented horn antenna, (b) finned horn antenna and (c) multimode horn antenna



Helical Antenna

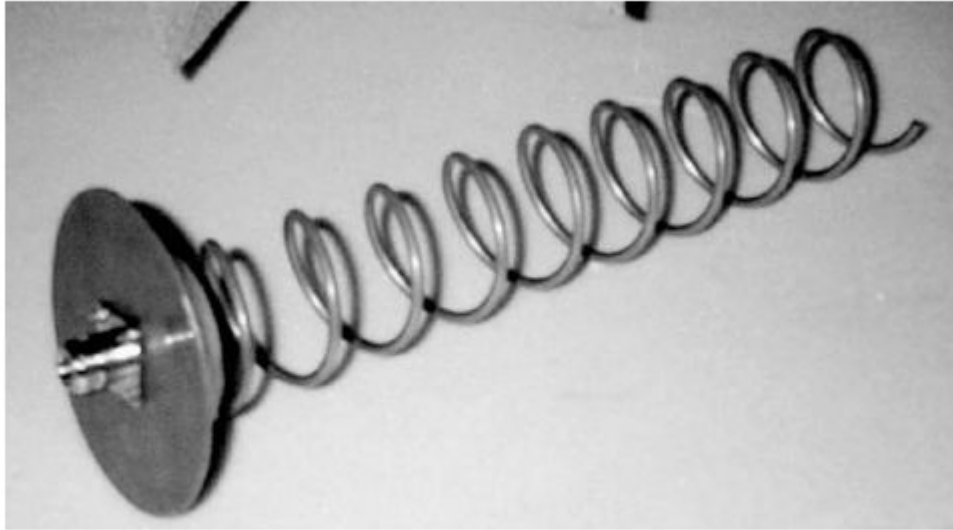


Figure 4.36 Typical helical antenna (Fabricated by High Gain Antenna Co. Ltd. Reproduced by permission of Hak-Keun Choi)



Lens Antenna

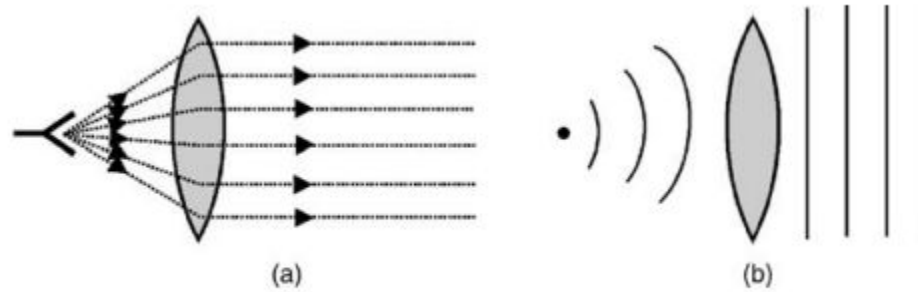


Figure 4.38 (a) Principle of operation of a lens antenna. (b) Wave diagram of a lens antenna



Phased Array Antenna

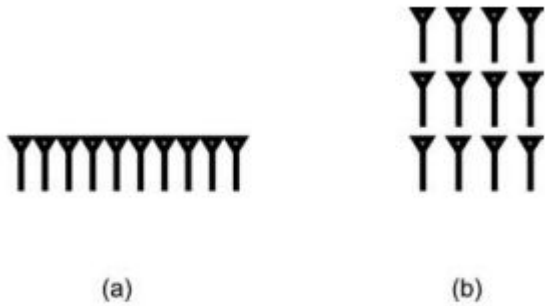


Figure 4.40 (a) Linear and (b) planar phased array ante

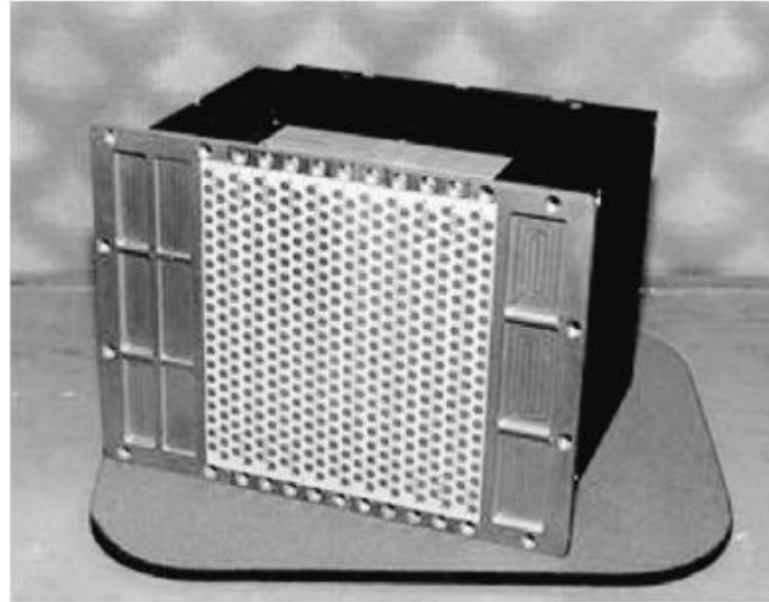


Figure 4.41 Photograph of a typical phased array antenna structure (Courtesy: NASA)